Design thinking educator workbook

COURSE OVERVIEW

The design thinking series of courses is designed to help you explore digital creativity in your classroom.

Course 1 - Introduction to design thinking. This course introduces design thinking as a framework to explore digital creativity in the classroom. It encourages learners to think about the challenges people experience and to develop solutions to help them.

Course 2 - Applying design thinking. This course shows how design thinking can help when tackling creative challenges with your learners. It uses examples aligned with the United Nation's Global Goals, to bring the content to life.

Design thinking framework:

EMPATHISE → DEFINE → IDEATE → PROTOTYPE → TEST



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USEFUL DEFINITIONS

The design thinking series of courses introduces some vocabulary that may be new to your class. We have collated definitions below for your reference.

vocabulary	definition
Design thinking	A human-centred approach to problem solving.
Empathise	To understand the needs and perspectives of others - in the Design thinking context it means having a deep understanding of the problems and realities of the people you are designing for.
Define	To state exactly what the scope of something is or decide the boundaries – in the Design thinking context learners identify which problem they would like to solve or which area they would like to focus on.
Ideate	To form an idea – in the Design thinking context it is a creative process where learners generate ideas for a variety of solutions.
Prototype	A first version of something - in the Design thinking context it refers to a simple model built to test ideas and changes.
Iteration	The repetition of a process – in the Design thinking context it refers to continuously improving a concept, design or product. A new version of something can be described as a new iteration.
Collaboration	The action of working with someone to produce something.
Perspective	A point of view or a particular attitude towards something.
Innovative	Something that uses new methods or ideas.

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USING DESIGN THINKING COURSES IN YOUR CLASSROOM

The design thinking series of courses are designed for you to replicate the learning with your students. Here is an outline of the key elements to consider when adapting the learning into a series of lessons. The courses are supported by a student workbook, with worksheets for each stage of the design thinking framework, including a choice of templates for different approaches.

Activity breakdown for each element of design thinking:

1 – EMPATHISE

In this stage, learners focus on understanding the needs and perspectives of others.

KEY STEPS	USEFUL TOOLS / CONTEXTS
1. Think about who the learners are designing for. 2. Explore what assumptions & bias are, and why they can be problematic. 3. Decide on useful research methods (see tools). 4. Learners use research to identify actions, thoughts & feelings of users they are designing for.	**Observations** – learn things people may not say aloud. **Interviews** – explore what's not easy to see and deepen understanding, validate information, and gain new insights. **Surveys** – gather large amounts of information. **Storyboards** – helpful when you can't speak directly to users, and in contexts when time is passing.

2 – DEFINE

In this stage, learners identify which area they would like to focus on.

KEY STEPS	USEFUL TOOLS / CONTEXTS
1. Bring together research. 2. Identify key findings/problems. 3. Group by theme. 4. Define core problem. 5. Decide which statements to focus on. 6. Create personas.	**Lists or tables** – decide on a common problem to work on together or to allocate different problems to different groups. **Highlighting or colour-coding tools** – categorise or prioritise. **Mind maps** – make connections between things. **Problem statement template** – define a problem. **Voting systems** – prioritise and make decisions. **Persona templates** – remind learners who they are designing for, link to literacy skills.

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3 – IDEATE

In this stage, learners generate ideas for a variety of solutions.

KEY STEPS	USEFUL TOOLS / CONTEXTS
1. Generate – come up with lots of ideas for solutions. 2. Synthesise – build a short list of unique ideas. 3. Prioritise – decide which ideas to take forward.	**Prioritisation matrix template** – visualise importance to help make decisions. **Dot voting** – visualise what idea is most popular.

4 – PROTOTYPE

In this stage, learners use simple models to refine their ideas.

KEY STEPS	USEFUL TOOLS / CONTEXTS
1. Take ideas and develop further. 2. Visualise ideas. 3. Build into a prototype. 4. Experiment with prototypes. 5. Make changes based on experimentation. 6. Choose 1 or 2 prototypes for robust testing.	**Sketches** – get ideas down quickly in a visual way. **Storyboards** – show use of something over time, with different users and in different contexts. **Models/modelling** – visualise and test ideas, aid problem solving, demonstrate how different components interact together and how users interact with the design.

5 – TEST

In this stage, learners test their idea to make sure it works to solve the problem.

KEY STEPS	USEFUL TOOLS / CONTEXTS
1. Take prototype and test to see if it solves the problem. 2. Share with people from the 'Empathise' stage. 3. Get feedback. 4. Observe people using the product/solution.	**Presentations** – share detailed ideas. **Interviews** – allow open-ended questions to be asked, and detailed opinions to be shared. **Note taking tools** – record observations. **Recording tools** – ensure information is not missed and can be reviewed at later date. **Testing worksheet** – ensure key factors are considered.

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PRESENTATION & REFLECTION	
In this stage, learners share their ideas and what they learnt.	
KEY STEPS	**USEFUL TOOLS / CONTEXTS**
1. Reflect on the solution and if it solved the problem. 2. Evaluate the solution using the 4 'L's. 3. Think about the audience and plan a presentation.	**Note taking tools** – to record reflections and evaluations. **Presentation software** – to create slides and videos.

Opportunities for wider learning outcomes at each stage of design thinking:

Skills	Empathise	Define	Ideate	Prototype	Test	Present
Thinking about others	√	√	√	√	√	
Creativity & imagination			√	√		√
Practical skills & hands-on experimentation				√	√	
Critical thinking including analysis & reflection		√	√	√	√	
Decision making		√	√	√	√	
Resilience				√	√	
Understanding how to give, accept and act on feedback	√			√	√	
Communication skills	√	√	√	√	√	√
Collaboration and team working	√	√	√	√	√	√

AFTER THIS SERIES OF COURSES

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